Prefix Sum

In Computer Science a prefix is a contiguous subarray that starts at very beginning of the array (index = 0) and extends upto some specific index i.

So in a prefix sum problem, the "prefix Sum" simply means the running total or cumulative sum of all the elements from the start of the array upto the current position.

index :	0	1	2	3
original Array :	5	2	8	1
	5	(5+2)	(5+2+8)	(5+2+8+1)
Prefix Sum Array :	5	7	15	16

mathematically the formula can be written :-

$$\boxed{prefix[i] = prefix[i-1] + arr[i]}$$

The ultimate power of computing these prefixes is that it allows you to find the sum of any random middle section (sub array) of that array in a constant time $O(1)$ without using any loop to re-add the numbers.

if you want the sum of elements from index L to R you take the prefix sum upto R and subtract the prefix sum just before L.

$$\boxed{sum(L,R) = prefix[R] - prefix[L-1]}$$

0	1	2	3
5	2	8	1

5 7 15

$$Sum(1,2) = 2+8 = 10$$

$$prefix(2) - prefix(0)$$

$$= 15 - 5$$

$$= 10$$

Postfix Sum (most commonly known as Suffix Sum)

* The exact opposite of Prefix Sum.

Instead of adding elements from the beginning you add elements starting from the very end of the array (index n−1). and work your way backward to the current position.

what is the suffix sum array :- A new array each index stores the total sum of all elements from that index to the end of the array.

$$\boxed{suffix[i] = suffix[i+1] + arr[i]}$$

0	1	2	3
5	2	8	1

Actual Array

5+2+8+1	2+8+1	8+1	1

16	11	9	1

$\boxed{\text{Suffix Array.}}$

or

$\boxed{\text{Postfix Array.}}$